MADHUMITHA KUMAR

SENIOR MACHINE LEARNING ENGINEER

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- ★ 5+ years of hands-on experience in applied data science, ML/DL including 2 years in generative AI (GenAI) designing and delivering enterprise-level AI solutions across industries in Real Estate, Government, Healthcare, and Energy, optimizing sales, payments, supply chain, operations, and customer service workflows.
- ★ Established expertise in a range of ML domains- NLP/NLU [Topic Modeling, Document Analysis, QA], Image Processing [Object Detection, OCR], Audio Processing [Speech-to-Text], Generative AI [LLM Inference, Optimization and Evaluation, VectorDBs, Search & Retrieval, RAG, Agentic Workflows, Guardrails], Forecasting, Recommendation Systems and Ranking.
- ★ Proficient in writing production-quality Python and SQL code, harnessing Spark and Cloud environments, utilizing containerization for packaging AI applications, designing robust AI architectures/systems, implementing cutting-edge algorithms and identifying new AI business opportunities.
- ★ Demonstrated success in operationalizing AI/data science products at scale to production and implementing techniques informed by latest research and state-of-the-art technologies.
- ★ Skilled in building end-to-end ML pipelines, prompt engineering for text-based LLMs, implementing MLOps/LLMOps, optimizing model architectures, API integrations and CI/CD development.

PROFESSIONAL EXPERIENCE

PwC Tech & Transformation, Senior Machine Learning Engineer — **January 2024 – Present**

- ❖ Enterprise AI in Healthcare:
  - ➢ Led the design, development, and operationalization of an enterprise-scale LLM application for a Fortune 500 healthcare payer accelerating provider contract processing amid growing acquisitions.
  - ➢ Orchestrated end-to-end MLops by implementing Azure PromptFlow, enabling model storage, versioning, and real-time monitoring for scalable, reliable, and maintainable ML deployments across the enterprise.
  - ➢ Collaborated cross-functionally with UI/UX, DevSecOps, Engineering, and Business teams to integrate ML capabilities into operational workflows and refine product features in an agile, iterative environment.
  - ➢ Architected and maintained a framework for ML output evaluation and automated testing, establishing end-to-end QA pipelines that supported continuous integration and reliable performance in production.
  - ➢ Performed in-depth code reviews on Databricks for AI engineers, ensuring best practices in ML engineering, scalability, and production-readiness on Azure Cloud to maintain robust, high-quality code.
  - ➢ Led the fine-tuning of few-shot learning prompts using Retrieval-Augmented Generation (RAG) to accurately contextualize and extract key attributes, automating the contract configuration process and improving operational efficiency.
- ❖ Healthcare Claim Edits GenAI Prototype:
  - ➢ Developed a GenAI demo application for automated claim edit generation from PDF documents, leveraging Databricks Delta tables for structured data storage, Llama 3.170B-instruct & GPT-4-o mini LLMs on Databricks for real-time inference, and Streamlit (through Databricks Apps) for a user-friendly UI.
  - ➢ Implemented Databricks SQL, endpoints, and Apps with end-to-end Databricks infrastructure to establish a streamlined data pipeline and real-time inference environment, ensuring seamless integration and high responsiveness.
- ❖ Real Estate MLOps Architecture:
  - ➢ Co-architected an enterprise-grade MLOps infrastructure for a leading real estate company and operationalized the first ML model (customer propensity classification with CatBoost algorithm) from experimentation to production on AWS for batch-inference.
  - ➢ Engineered reusable, scalable CI/CD and MLOps pipeline templates for model monitoring and governance, supporting batch and real-time inference, multi-model applications, automated retraining, automated feature store ingestion and scheduling to facilitate productionalization applicability across different use cases.
- ❖ Retail Pharmacy ML Architecture:
  - ➢ Designed and integrated SOTA ML algorithms into scalable, Azure architectures for a Fortune 100 retail pharmacy, addressing critical healthcare needs through advanced ML solutions such as pill identification (Object Detection, CV), customer support (IVR, agent assistance), Rx processing (OCR, information extraction) and insurance claim rejections processing (NLP, classification, LLM) driving efficiency across healthcare workflows.
  - ➢ Enabled "Next-Best Actions" recommendations for customer service agents in Agent Assist AI solution with RAG-based LLM application, enhancing customer experience through proactive support.

- ➢ Provided data-driven impact assessments to quantify potential improvements and aid strategic investment decisions, ensuring AI solutions aligned with business goals and added measurable value.

PwC Labs, Senior Data Scientist **August 2022 – December 2023**

- **DocQA GenAI Application:**
  - ➢ Led advanced research and experimentation with GraphRAG techniques to enhance PwC's RAG-based knowledge management platform, positioning it as a competitive alternative to Salesforce Einstein and improving retrieval accuracy for Fortune 500 clients.
  - ➢ Developed a Proof of Concept (PoC) to evaluate and benchmark GraphRAG approaches, demonstrating performance improvements over VectorRAG and presenting findings to key stakeholders to guide strategic decisions on model enhancements.
  - ➢ Collaborated closely with business and engineering teams to refine feature requirements, translating business objectives into technical specifications for scalable knowledge retrieval solutions.
  - ➢ Enhanced the FAISS vector index pipeline to support diverse document types, including OCR for images and scanned documents, enabling comprehensive knowledge coverage and improving the platform’s adaptability.
  - ➢ Engineered an API integration for real-time transcription from Amazon Connect to the application backend, facilitating seamless, real-time document search and retrieval for enhanced knowledge accessibility.
  - ➢ Drove experimentation with agent-based retrieval approaches to improve contextual understanding within the knowledge management system, continuously enhancing accuracy and responsiveness for real-time information needs.
- **Contact Center AI System:**
  - ➢ Led optimization of PwC’s flagship contact center AI solution deployed across three major clients, achieving significant scalability, performance, and latency improvements through fine-tuning ML models, pipeline enhancements, and backend code optimization.
  - ➢ Refactored source code with a modular, reusable design that enabled rapid prototyping and customization for specific client needs, empowering cross-functional teams to efficiently deliver tailored AI solutions while minimizing technical debt and enhancing system flexibility.
  - ➢ Expanded the solution’s multimodal data handling to support diverse sources such as audio, chat transcripts, emails, and IT tickets, allowing seamless integration of this data into the ML pipeline and broadening the AI solution's applicability for customer service use cases.
  - ➢ Integrated cutting-edge GenAI capabilities using OpenAI models (GPT-3.5, GPT-4) with custom prompt engineering to enable semi-supervised labeling, summarization, and classification, enhancing automated support functions.
  - ➢ Utilized Azure AI and Spark for high-performance data processing, including Spark and GPU-accelerated RAPIDS libraries for batch processing of large datasets, achieving scalable email data ingestion and analysis.
  - ➢ Employed MLflow and Databricks for model tracking and deployment, ensuring consistent performance monitoring and reliable model serving across demo and production environments.
  - ➢ Integrated Azure AI APIs with Spark, Hive, and GPU-accelerated RAPIDS libraries to enable high-performance data processing and reliable inference, including efficient data storage, joining, filtering, and analysis at scale. Leveraged Spark and MapReduce for batch processing of large datasets, achieving scalable email data ingestion and analysis across millions of records.
  - ➢ Developed and deployed two PoC applications using KNative framework for automated ticket resolution and email routing, leveraging containerized environments (Docker, Kubernetes) and CI/CD automation to ensure reliability, scalability, seamless code updates and efficient delivery.
  - ➢ Managed Git repositories and reviewed feature pull requests, mentoring junior data scientists and operationalizing all modules into production-ready Python scripts from notebooks to streamline experimentation and maintain high code standards.
  - ➢ Implemented robust software engineering practices including unit testing, design documentation, and thorough code reviews to ensure maintainable, high-quality code and smooth deployment of modules.

CGI Federal Consulting | SSA Analytics CoE, Data Scientist **January 2020 – July 2022**

- Developed a semi-supervised deep learning classification model on TF framework and small language models to predict district and circuit court outcomes, optimizing litigation strategy and aiding district attorney capacity planning with data-driven insights.
- Engineered and deployed ETL pipelines to pull structured data from multiple relational and NoSQL databases, capturing complex adjudication processes for social security claims in an efficient, automated workflow.
- Built a scalable OCR pipeline for feature extraction from court documents and leveraged PySpark to query and transform large-scale medical data stored in Hadoop clusters, converting Parquet files into Pandas DataFrames for analysis.
- Performed feature importance analysis using advanced statistical techniques, including multivariate regression, PCA, and Chi-Square tests, identifying and proposing new features that significantly improved model accuracy.
- Contributed to the development of a secure internal deep learning package in Python, enabling robust analysis of sensitive unstructured data with NLP methods such as embeddings and clustering for advanced document classification and categorization.
- Demonstrated best practices in model validation and deployment within a secure, continuous integration environment, ensuring that ML solutions met high standards of accuracy, reliability, and scalability.
- Built complex, integrated datasets using SQL and PySpark, powering BI dashboards and assembling training datasets for ML models, aligning data engineering efforts with robust ML and BI development needs.
- Optimized ETL pipelines through enhanced SQL joins and aggregations, achieving a 30% reduction in data refresh times, and led a team of data engineers in developing efficient, scalable ETL processes for analytics and reporting.
- Conducted A/B analysis to evaluate new policy impacts, using ANOVA and linear regression to compare pre- and post-policy populations and identify statistically significant differences, informing congressional reporting with data-backed insights.
- Enhanced workload forecasting models by implementing ensemble algorithms including ARIMA, ETS, Holt-Winters, and neural networks to accurately predict district-level workloads, optimizing resource allocation and attorney staffing across regions.

SKILLS STACK

API Management: Swagger Docs, Postman, cURL
Agentic AI: Autogen, Swarm
CI/CD Tools: GitHub Actions, Azure Pipelines, Jenkins
Data Engineering and ETL: Hadoop, Hive, Spark, Azure Synapse, AWS Glue
Containerization: Docker, Kubernetes, AWS ECR
Forecasting: SAS, FB Prophet
IaC: Terraform, AWS CloudFormation
LLMOps: Langsmith, Langfuse, Prompt Flow
Graph Data: Neo4j, NebulaGraph
GenAI Frameworks: LangChain, Llama Index, Ollama
ML Frameworks: Tensorflow, Keras, PyTorch, Hugging Face, RAPIDS, Scikit-Learn
Managed ML Development: AWS Sagemaker, Azure ML
ML Techniques: Ensemble, Boosting, Attention-based, Recurrence-based, Convolution-based
MLOps: Databricks, MLflow, Weights & Biases, Sagemaker
Programming: Python, Bash/Shell, SQL, SAS, PySpark, R
Version Control: Git, BitBucket, JIRA, Azure DevOps
Relational DBs: DB2, PostgreSQL, SQL Server, AWS Redshift, AWS Athena, DynamoDB, BigQuery, MySQL
Vector Databases: FAISS, Azure AI Search
LLM Evaluation: RAGAS, Evals, DeepEval, LLM-as-a-judge, Nemo Guardrails
ML/Data Application Development: Streamlit, Flask, Django, FastAPI, Gradio, Replit

ADDITIONAL SKILLS

Design/Wireframing/UI: Figma, Visio
Web Development: CSS, HTML, JavaScript
BI Engineering: Tableau, CRMAnalytics, Power BI
Visualization: D3.js, Matplotlib, Seaborn, Plotly

CERTIFICATIONS

Azure AI Engineer Associate, Microsoft, 12-2024
Azure AI Fundamentals, Microsoft, 03-2024

Data Science with Sagemaker, AWS, 08-2021
Architecting on AWS, AWS, 06-2021
Forecasting using SAS Forecast Studio, SAS, 10-2020

PROFESSIONAL ACTIVITIES

- PwC's Women in AI; Eminence

EDUCATION

General Assembly, Immersive Data Science Bootcamp, 2019

- *Relevant Focus:* Data Structures and Algorithms, Data Modeling, Python Programming

Georgetown University, Master's in Communication, Culture and Technology, 2019

- *Relevant Focus:* AI & Autonomous Systems; Fundamentals of IT; AI & The Future of Work; Cognitive Technologies, Econometrics, Introduction to Python & R

University of Madras, Bachelor of Science in Visual Communication, 2017

- *Relevant Focus:* Computer Modeling & 3D Animation (Maya); Website Development & Design